

## سوالات تحلیلی

۱. در زمان ارائه پردازنده 8086 چرا سه باس آدرس، داده ورودی و داده خروجی در پایه های تراشه با هم ترکیب شده‌اند؟ هنگام اتصال به حافظه خارجی چگونه این سه گذرگاه از هم تفکیک میشوند؟

**پاسخ:** با ترکیب کردن باس دیتا و باس 16 بیت اول آدرس، تعداد اتصالات فیزیکی و پین های مورد نیاز کاهش پیدا میکند. این ترکیب با استفاده از سیگنال های کنترلی مانند:

ALE: این سیگنال پورت های  $AD_0-AD_{15}$  را با کمک latch پایین به  $A_0-A_{15}$  و  $D_0-D_{15}$  demultiplex می‌کند.

DEN: این سیگنال به transceiver اعلام میکند که آماده تبادل داده است یا نه.

DT/R: این سیگنال برای کنترل گردش داده استفاده میشود. به این صورت که اگر high باشد نشان میدهد که دیتا در حال ارسال و اگر low باشد دیتا در حال دریافت است.

تفکیک میشود.

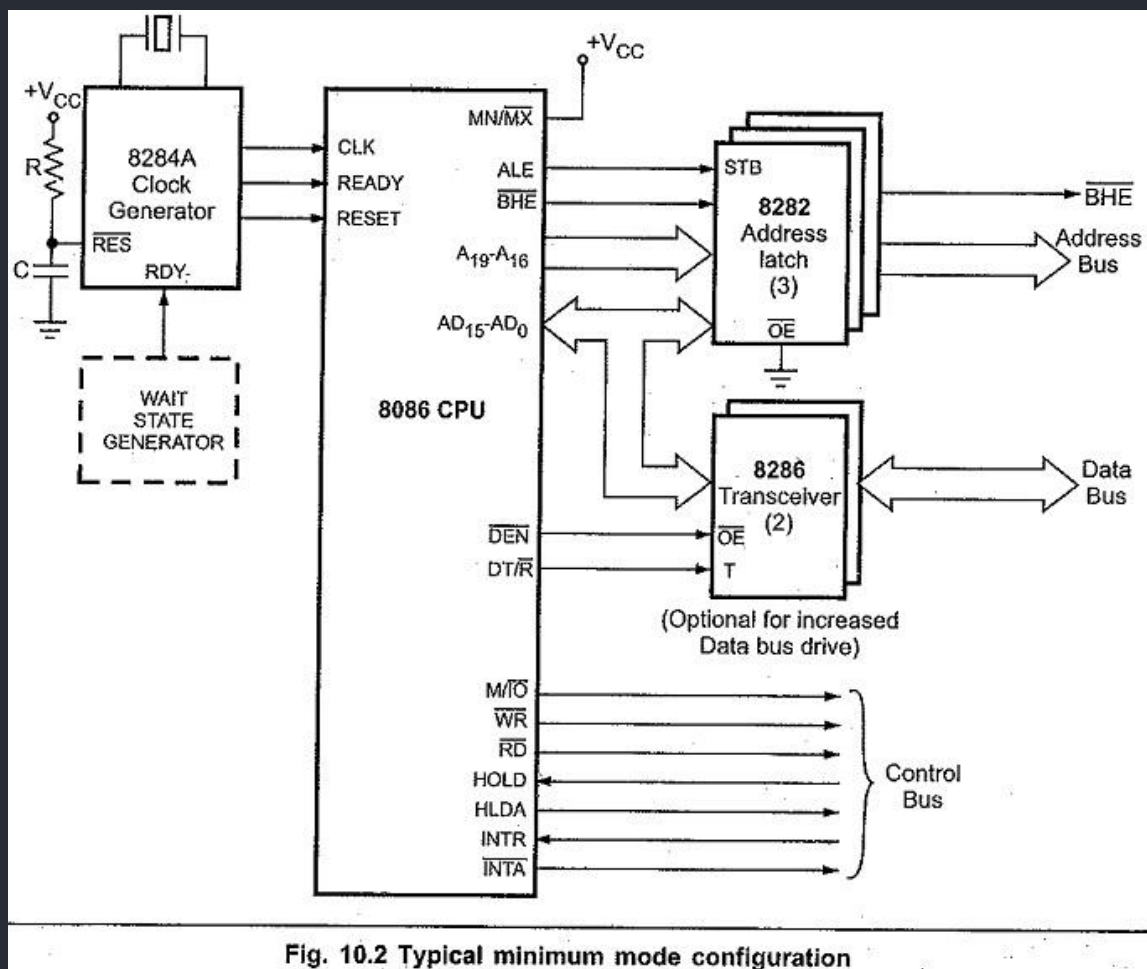
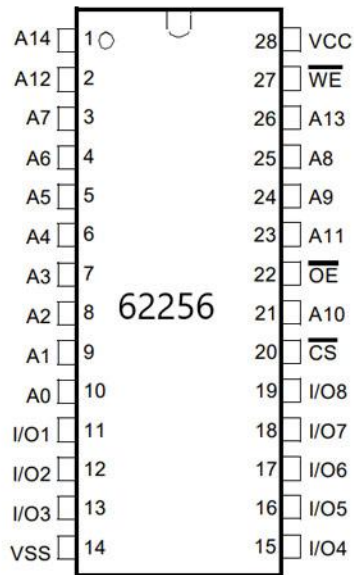


Fig. 10.2 Typical minimum mode configuration

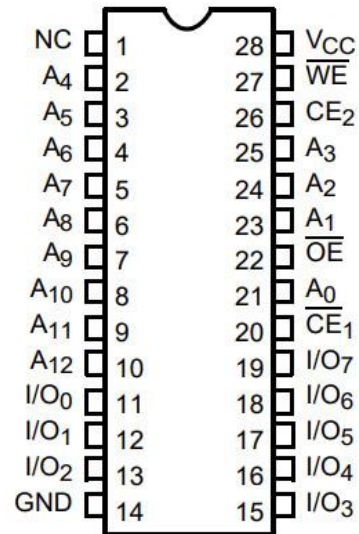
۲. با بررسی دیتاشیت تراشه های 6264 و 62256 و 2764 و 27128 سیگنالهای آدرس، داده و کنترلی هرکدام را مشخص کنید. عملکرد کلی تراشه و وظیفه هرکدام از پایه ها را شرح دهید.

**پاسخ:** هرکدام از تراشه های 6264 و 62256 یک حافظه RAM با 8Kbx8bit و 32Kbx8bit فضا اند و هر دو 28 پین دارند.

## 62256 Pinout



## 6264 Pinout



|               |                     |  |               |                     |
|---------------|---------------------|--|---------------|---------------------|
| $A_0-A_{14}$  | Address Inputs      |  | $A_0-A_{12}$  | Address Inputs      |
| $I/O_1-I/O_8$ | Data inputs/outputs |  | $I/O_0-I/O_7$ | Data inputs/outputs |
| WE            | Write Enable        |  | WE            | Write Enable        |
| OE            | Output Enable       |  | OE            | Output Enable       |
| CS            | Chip Select         |  | $CE_1, CE_2$  | Chip Enable         |
|               |                     |  | NC            | No Connection       |
| VCC           | Power Supply        |  | VCC           | Power Supply        |
| VSS           | Ground              |  | GND           | Ground              |

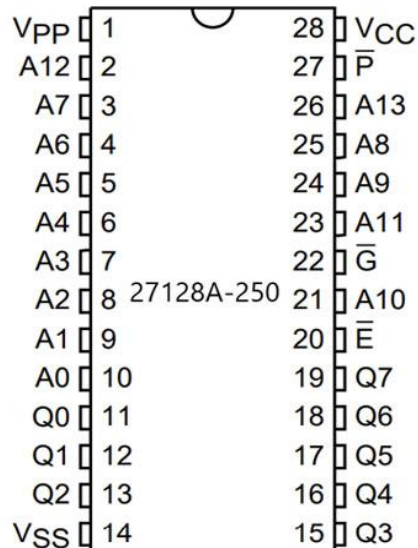
## FUNCTIONAL DESCRIPTION

| $\overline{CS}$ | $\overline{OE}$ | $\overline{WE}$ | I/O Pin | Mode            | Power   |
|-----------------|-----------------|-----------------|---------|-----------------|---------|
| H               | X               | X               | High-Z  | Deselected      | Standby |
| L               | H               | H               | High-Z  | Output Disabled | Active  |
| L               | L               | H               | Dout    | Read            | Active  |
| L               | X               | L               | Din     | Write           | Active  |

1. X means don't care

همچنین هر کدام از تراشه های 2764 و 27128 یک حافظه EPROM با 8Kbx8bit و 16Kbx8bit فضا اند و هر دو 28 پین دارند. کلاhek بی رنگ آنها این اجازه را به کاربر می‌دهد که با در معرض قرار دادن آنها در مقابل اشعه فرابنفش بیت پترن را پاک کند.

## 27128A-250 Pinout



## 2764-250 Pinout

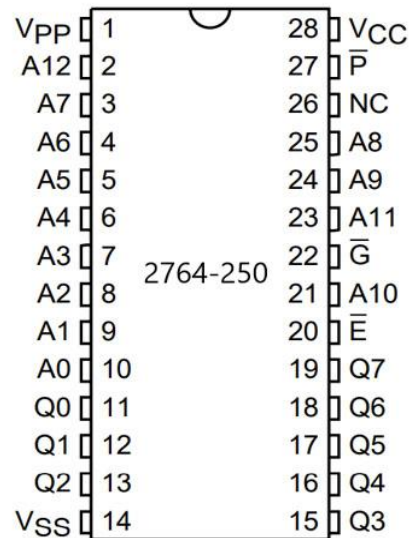


Table 3. Operating Modes

| Mode                 | $\overline{E}$  | $\overline{G}$  | $\overline{P}$        | A9              | Vpp             | Q0 - Q7   |
|----------------------|-----------------|-----------------|-----------------------|-----------------|-----------------|-----------|
| Read                 | V <sub>IL</sub> | V <sub>IL</sub> | V <sub>IH</sub>       | X               | V <sub>CC</sub> | Data Out  |
| Output Disable       | V <sub>IL</sub> | V <sub>IH</sub> | V <sub>IH</sub>       | X               | V <sub>CC</sub> | Hi-Z      |
| Program              | V <sub>IL</sub> | V <sub>IH</sub> | V <sub>IL</sub> Pulse | X               | V <sub>PP</sub> | Data In   |
| Verify               | V <sub>IL</sub> | V <sub>IL</sub> | V <sub>IH</sub>       | X               | V <sub>PP</sub> | Data Out  |
| Program Inhibit      | V <sub>IH</sub> | X               | X                     | X               | V <sub>PP</sub> | Hi-Z      |
| Standby              | V <sub>IH</sub> | X               | X                     | X               | V <sub>CC</sub> | Hi-Z      |
| Electronic Signature | V <sub>IL</sub> | V <sub>IL</sub> | V <sub>IH</sub>       | V <sub>ID</sub> | V <sub>CC</sub> | Codes Out |

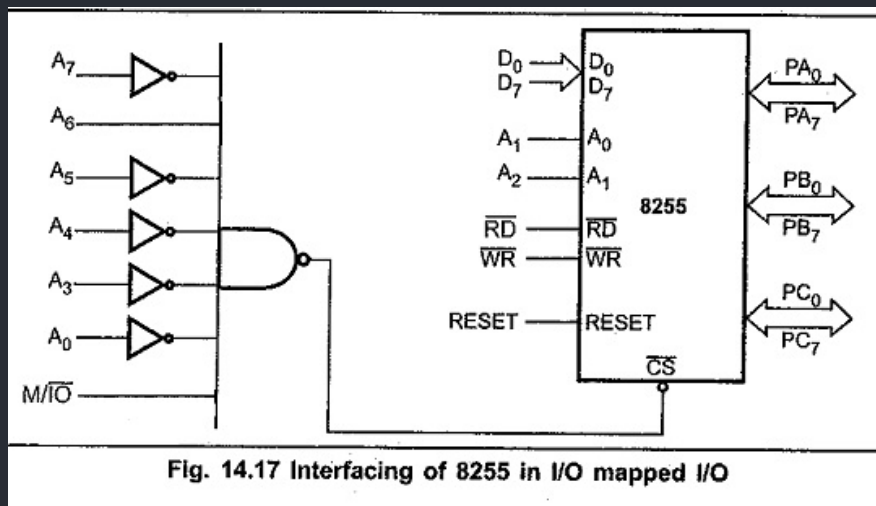
Note: X = V<sub>IH</sub> or V<sub>IL</sub>, V<sub>ID</sub> = 12V ± 0.5%.

|              |                    |  |              |                     |
|--------------|--------------------|--|--------------|---------------------|
| $A_0-A_{13}$ | Address Inputs     |  | $A_0-A_{12}$ | Address Inputs      |
| $Q_0-Q_7$    | Data outputs       |  | $Q_0-Q_7$    | Data outputs        |
| E            | Chip Enable        |  | E            | Chip Enable input   |
| G            | Output Enable      |  | G            | Output Enable input |
| P            | Program Pin        |  | P            | Program Pin         |
|              |                    |  | NC           | No Connection       |
| VPP          | Programming Supply |  | VPP          | Programming Supply  |
| VCC          | Power Supply       |  | VCC          | Power Supply        |
| VSS          | Ground             |  | VSS          | Ground              |

۳. فرض کنید قرار است از تراشه ۸۲۵۵ برای ارتباط پردازنده با دستگاهای ورودی / خروجی از طریق سه درگاه A, B و C آن استفاده شود.

الف. با به کارگیری دروازه NAND دیکد آدرس را طوری انجام دهید که آدرس درگاه A برابر 40H, درگاه B برابر 42H, درگاه C برابر 44H و کلمه کنترلی برابر 46H باشد.

پاسخ:



| Port             | A <sub>7</sub> | A <sub>6</sub> | A <sub>5</sub> | A <sub>4</sub> | A <sub>3</sub> | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | Address |
|------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------|
| Port A           | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 40H     |
| Port B           | 0              | 1              | 0              | 0              | 0              | 0              | 1              | 0              | 42H     |
| Port C           | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 0              | 44H     |
| Control Register | 0              | 1              | 0              | 0              | 0              | 1              | 1              | 0              | 46H     |

ب. decode فوق را با به کارگیری 74LS138 decoder انجام دهید.

پاسخ: خروجی دیکودر فوق active low است.

| Inputs |             |        |   |   | Outputs |    |    |    |    |    |    |    |
|--------|-------------|--------|---|---|---------|----|----|----|----|----|----|----|
| Enable |             | Select |   |   |         |    |    |    |    |    |    |    |
| G1     | G2 (Note 1) | C      | B | A | Y0      | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 |
| X      | H           | X      | X | X | H       | H  | H  | H  | H  | H  | H  | H  |
| L      | X           | X      | X | X | H       | H  | H  | H  | H  | H  | H  | H  |
| H      | L           | L      | L | L | L       | H  | H  | H  | H  | H  | H  | H  |
| H      | L           | L      | L | H | H       | L  | H  | H  | H  | H  | H  | H  |
| H      | L           | L      | H | L | H       | H  | L  | H  | H  | H  | H  | H  |
| H      | L           | L      | H | H | H       | H  | L  | H  | H  | H  | H  | H  |
| H      | L           | H      | L | L | H       | H  | H  | L  | H  | H  | H  | H  |
| H      | L           | H      | L | H | H       | H  | H  | H  | L  | H  | H  | H  |
| H      | L           | H      | H | L | H       | H  | H  | H  | H  | L  | H  | H  |
| H      | L           | H      | H | H | H       | H  | H  | H  | H  | H  | L  | H  |
| H      | L           | H      | H | H | H       | H  | H  | H  | H  | H  | H  | L  |

DM74LS138

DATA OUTPUTS

VCC

Y0 Y1 Y2 Y3 Y4 Y5 Y6

16 15 14 13 12 11 10 9

1 2 3 4 5 6 7 8

A B C G2A G2B G1 Y7 GND

SELECT

ENABLE

OUTPUT

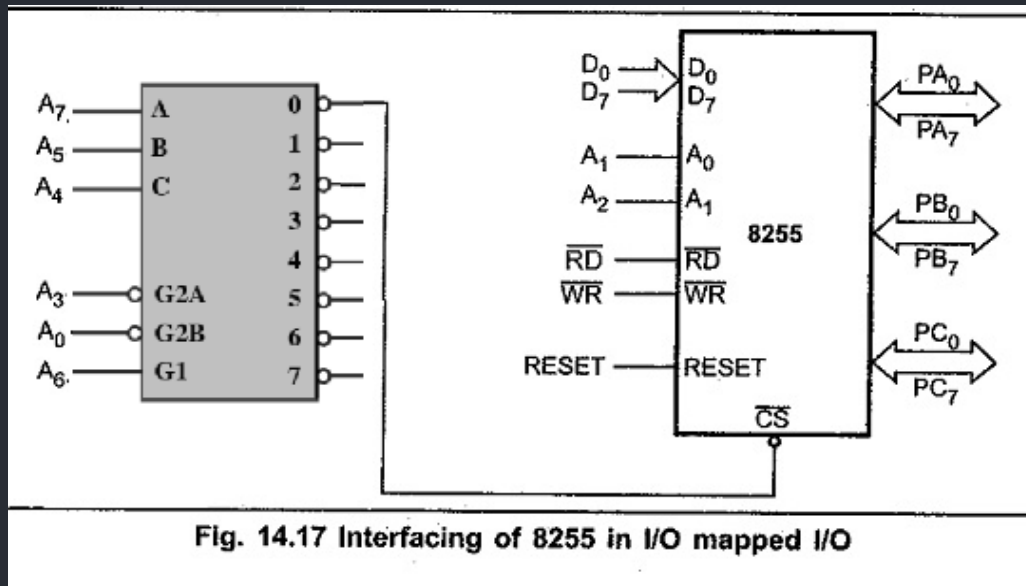
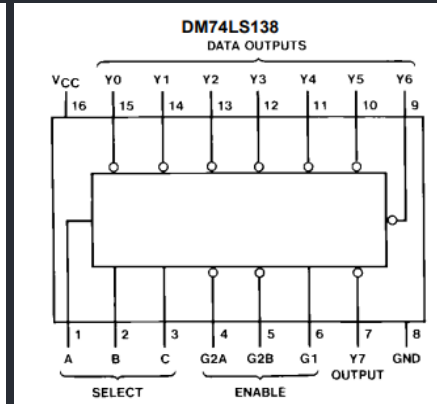
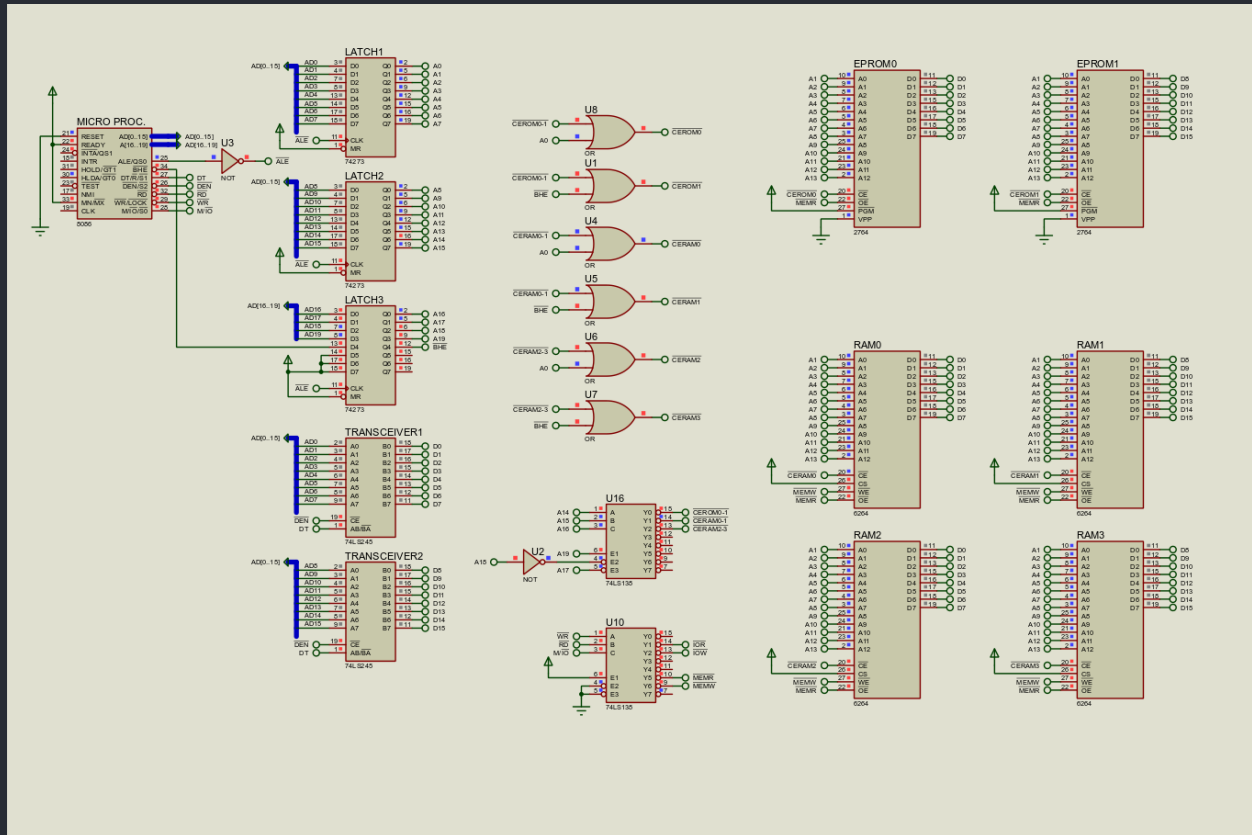


Fig. 14.17 Interfacing of 8255 in I/O mapped I/O

## گزارش دستور کار (الف):

### شمای مدار در پروتئوس:



مدار شامل 3 LATCH و 2 TRANSCEIVER به منظور جداسازی باس آدرس و دیتا است.

در دیکودر پایین صفحه سیگنال های MEMR و MEMW مشخص شده اند.

در دیکودر بالای آن آدرس هفتمین یک هشتم را مشخص میکند.  $0xC0000$

سیگنال های CE حافظه ها هم با استفاده از A0 و BHE تشکیل شده اند.

بازه ی حافظه ها

|                               |   |                               |
|-------------------------------|---|-------------------------------|
| EVEN                          | - | ODD                           |
| EPROM0 = 0xC0000: 2 :0xC3FFFE | - | EPROM1 = 0xC0001: 2 :0xC3FFFF |
| RAM0 = 0xC4000: 2 :0xC7FFFE   | - | RAM1 = 0xC4001: 2 :0xC7FFFF   |
| RAM2 = 0xC8000: 2 :0xCBFFFE   | - | RAM3 = 0xC8001: 2 :0xCBFFFF   |

## کد روتین KPEXT:

توضیحات کد کامنت شده است.

```

[01] Main.asm M X
Lab2-3 > Lab2 > [01] Main.asm
1  .MODEL SMALL
2  .STACK 64
3  .DATA
4      ROM DW 1100000000000000B
5      RAM DW 1100010000000000B
6      KPMN DB ?
7
8      M DB ?
9      N DB ?
10
11     r DB 01H
12     x_fact DB ? ; IS USED TO CALCULATE factorial of x
13     n_fact DW ?
14     r_fact DW ?
15     n_r_fact DW ?
16     comb DB ? ; ANSWER OF Combination
17     OUTER_LOOP_COUNT_TEMP DW ?
18 .CODE
19
20 MAIN    PROC FAR
21
22     CALL KPEXT
23
24 ENDLESS:
25     JMP ENDLESS
26 MAIN    ENDP
27

```

```
30 KPEXT PROC
31 READ_ROM:
32     MOV AX, @DATA
33     MOV DS, AX
34
35     MOV AX, ROM
36     MOV DS, AX
37
38     MOV DI, 00H
39     MOV CL, [DI] ; READ M
40     MOV CH, [DI+1] ; READ N
41
42 SAVE_DATA:
43     MOV AX, @DATA
44     MOV DS, AX
45
46     DEC CL
47     MOV M, CL
48     DEC CH
49     MOV N, CH
```



```

51 calculation:
52     MOV CX, 0000H; COUNTER = 0
53 L1:
54     MOV AX, @DATA
55     MOV DS, AX ; SET DS TO @DATA
56
57     CMP CL, M
58     JA L1_END ; FINISH WHEN COUNTER > M
59
60     MOV r, CL ; r = COUNTER
61
62     ; --- M! ---
63     MOV AH, M
64     MOV x_fact, AH ; x = M
65     MOV BL, AH
66     CALL FACT
67     MOV n_fact, AX
68     ; -----
69
70     ; --- r! ---
71     MOV DL, r
72     MOV x_fact, DL ; x = r
73     CALL FACT
74     MOV r_fact, AX
75     ; -----
76
77     ; --- (M-r)! ---
78     MOV DL, M ; DL = M
79     SUB DL, r ; DL = M - r
80     MOV x_fact, DL ; x = M - r
81     CALL FACT
82     MOV n_r_fact, AX
83     ; -----
84
85     ; --- M!/(r! (M-r)!) ---
86     MOV AX, n_fact
87     SUB DX, DX
88     DIV r_fact
89     SUB DX, DX
90     DIV n_r_fact
91     MOV comb, AL
92     ; -----
93
94
95     SUB BX, BX ; Clear AX
96     MOV BL, comb ; AX = comb
97

```

```

99  ▾ WRITE_RAM:
100      MOV AX, RAM
101      MOV DS, AX
102
103      MOV DI, CX
104      MOV [DI], BL ; WRITE COMBINATION IN RAM
105
106      INC CX ; COUNTER++
107      INC CX ; COUNTER++
108      LOOP L1 ; COUNTER--
109
110  ▾ L1_END:
111      SUB BX, BX
112      MOV BL, N
113      MOV DI, BX ; DI = N
114
115      MOV AX, RAM
116      MOV DS, AX
117
118      MOV BL, [DI] ; READ Nth COMBINATION
119
120      MOV AX, @DATA
121      MOV DS, AX
122
123      MOV KPMN, BL ; KPMN = M
124
125      RET
126
127      KPEXT      ENDP

```

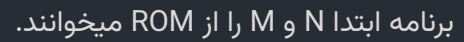
```

130  ✓ FACT PROC ; Saves x! IN (DX, AX)
131
132      MOV OUTER_LOOP_COUNT_TEMP, CX
133
134      MOV CX, 01H ; COUNTER = 1
135      MOV AX, 01H ; AX = 1
136      MOV DX, 0000H ; CLEAR DX
137
138  ✓ BACK:
139      MOV BH, 00H
140      MOV BL, x_fact
141      CMP CX, BX
142      JA FINISH ; FINISH WHEN COUNTER > x
143      MUL CX ; (DX, AX) = CX * AX
144      INC CX ; COUNTER++
145      INC CX ; COUNTER++
146      LOOP BACK ; COUNTER--
147
148
149  ✓ FINISH:
150      MOV CX, OUTER_LOOP_COUNT_TEMP
151      RET
152      ; THE ANSWER IS STORED IN (DX, AX)
153
154  ✓ FACT ENDP
155  /// END MAIN3

```

$M = 6$ 

سومین عدد خط ششم مثلث خیام پاسکال =  $0AH = 10D$

[illegible]



## گزارش دستور کار (ب):

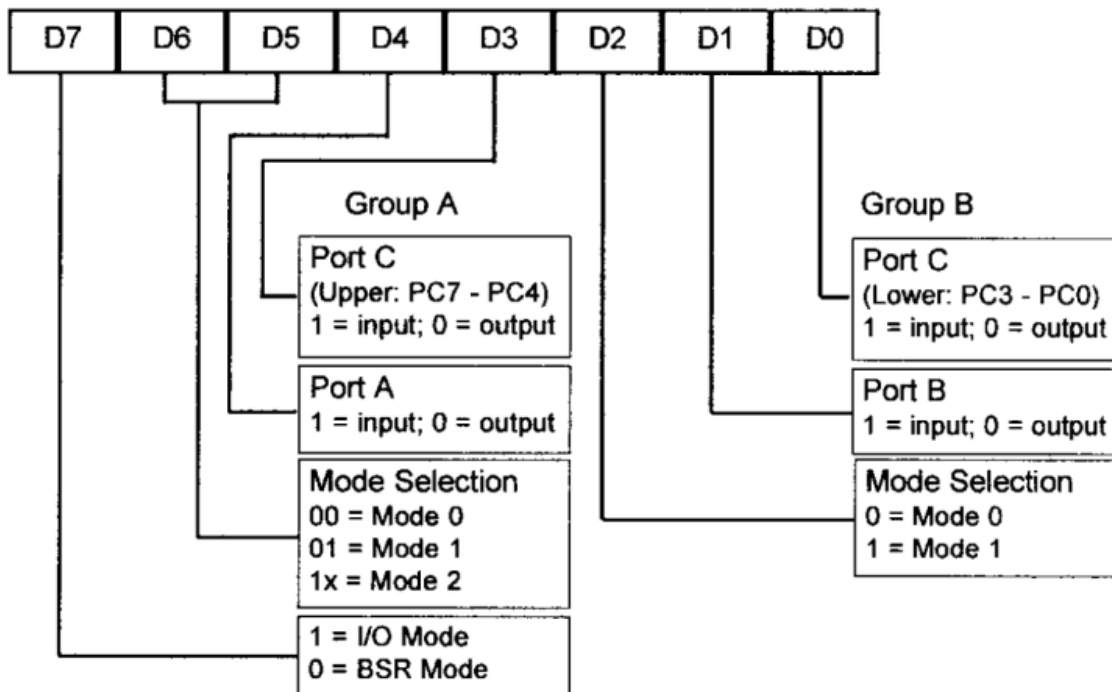
فضای آدرس چیپ 8255 به صورت زیر دیکود شده است:

I/O Map :

| Port / control Register | Address lines                                   | Address |
|-------------------------|-------------------------------------------------|---------|
|                         | $A_7$ $A_6$ $A_5$ $A_4$ $A_3$ $A_2$ $A_1$ $A_0$ |         |
| Port A                  | 0 0 0 0 0 0 0 0                                 | 00H     |
| Port B                  | 0 0 0 0 0 0 1 0                                 | 02H     |
| Port C                  | 0 0 0 0 0 1 0 0                                 | 04H     |
| Control register        | 0 0 0 0 0 1 1 0                                 | 06H     |

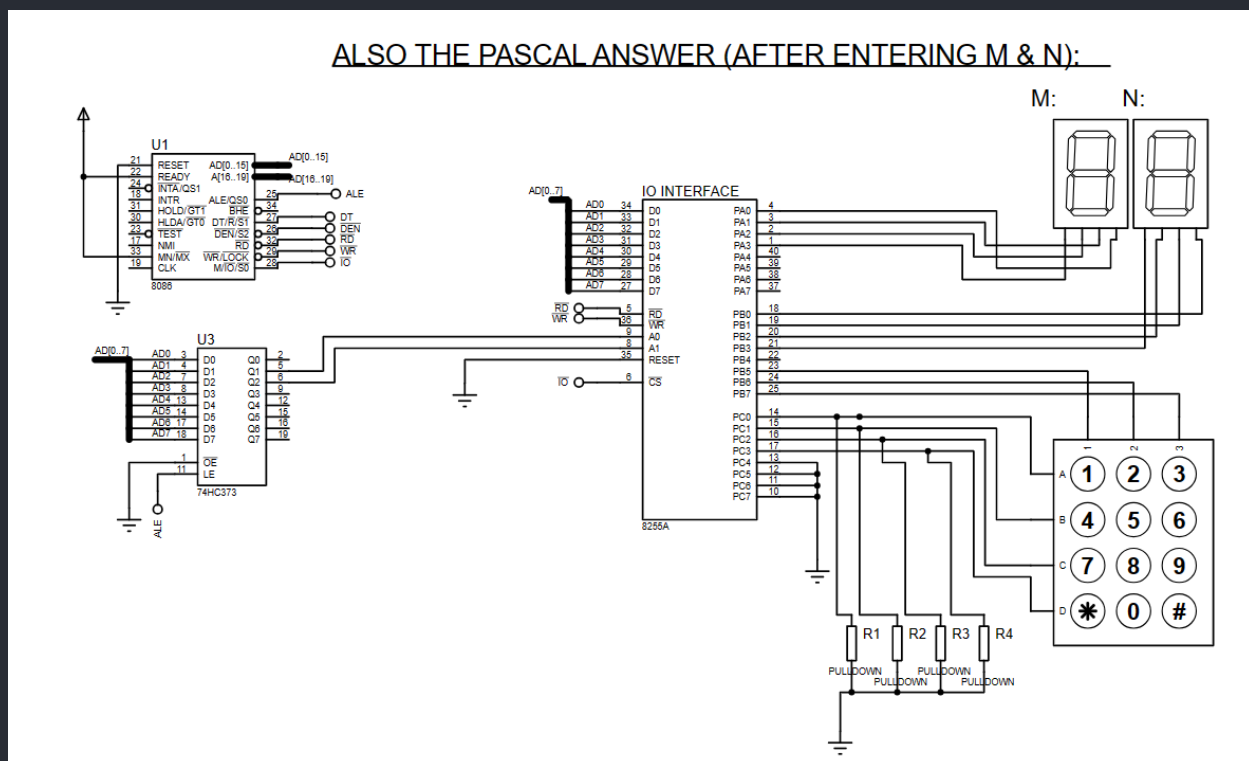
طبق:

## Mode Selection with Control Register



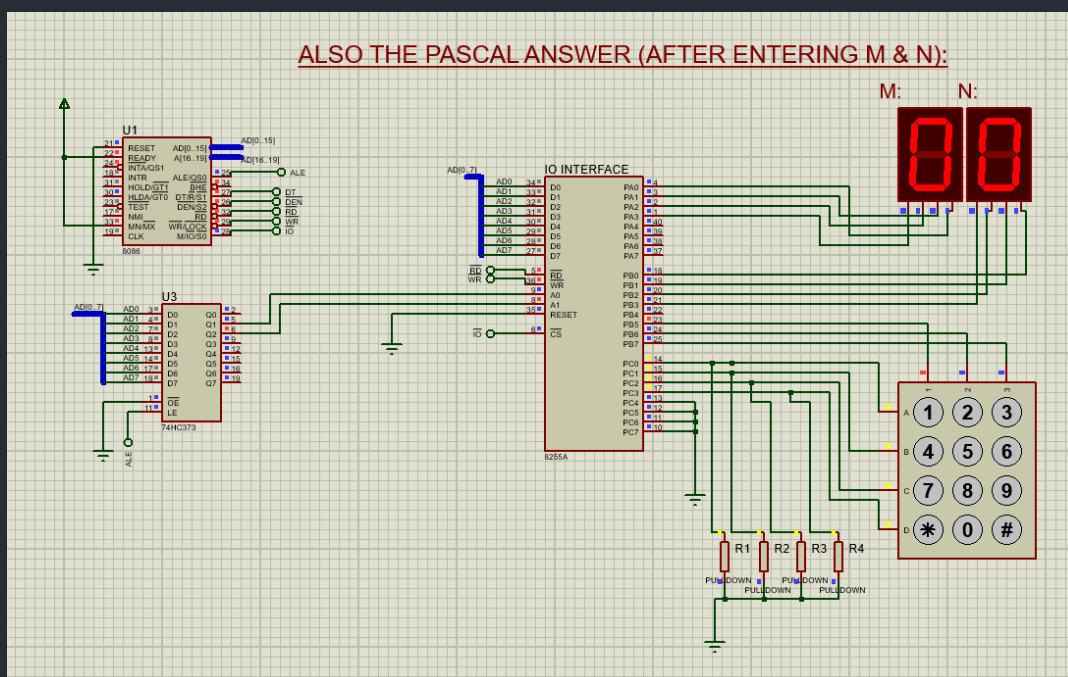
کنترل رجیستر این 8255 حاوی بیت های 10001001 است.

## شمای مدار در پروتئوس:

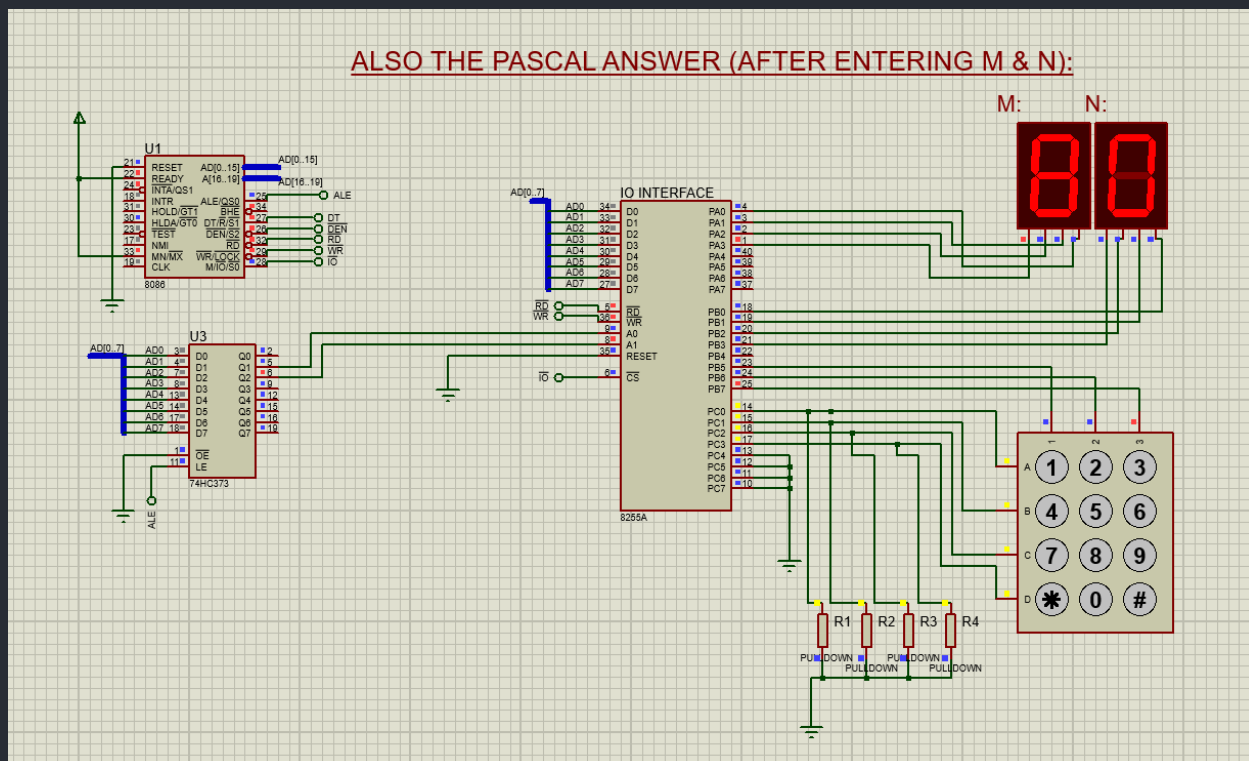


خروجی نمونه (۴امین عدد سطر ۱۸م):

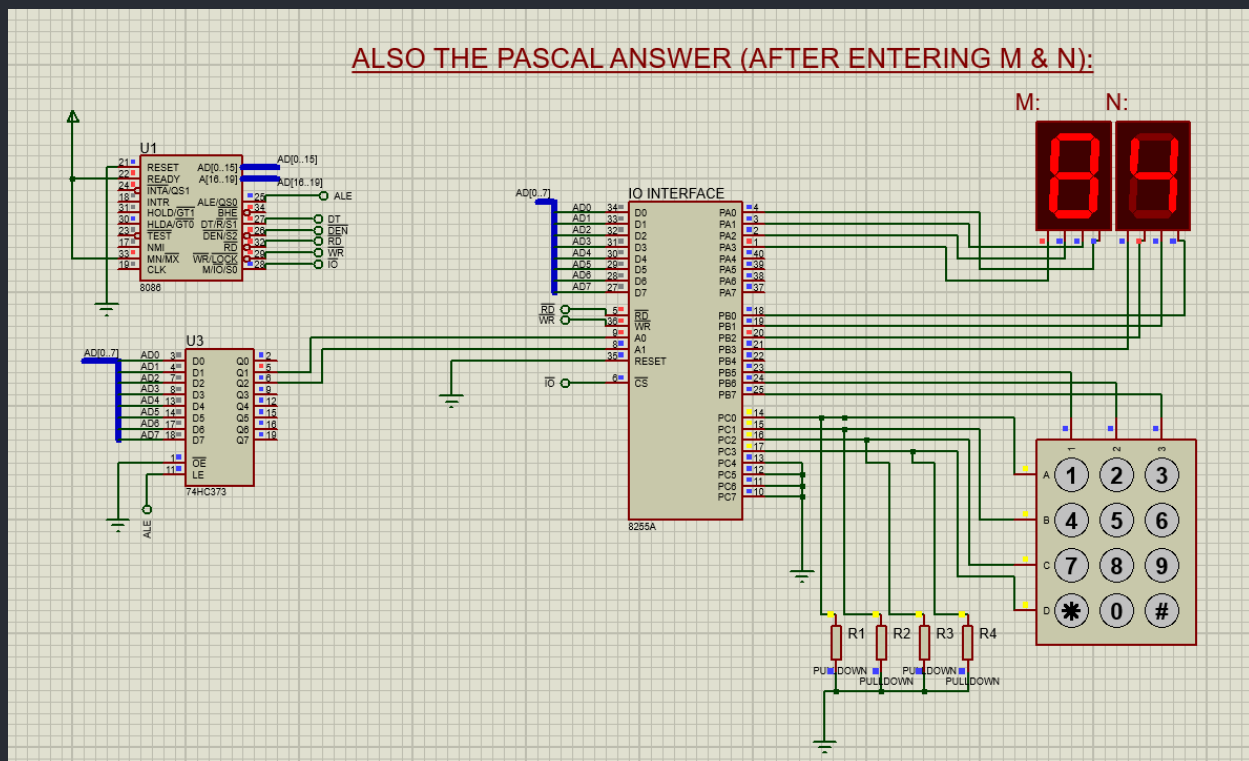
حالت اولیه:



وارد کردن M:

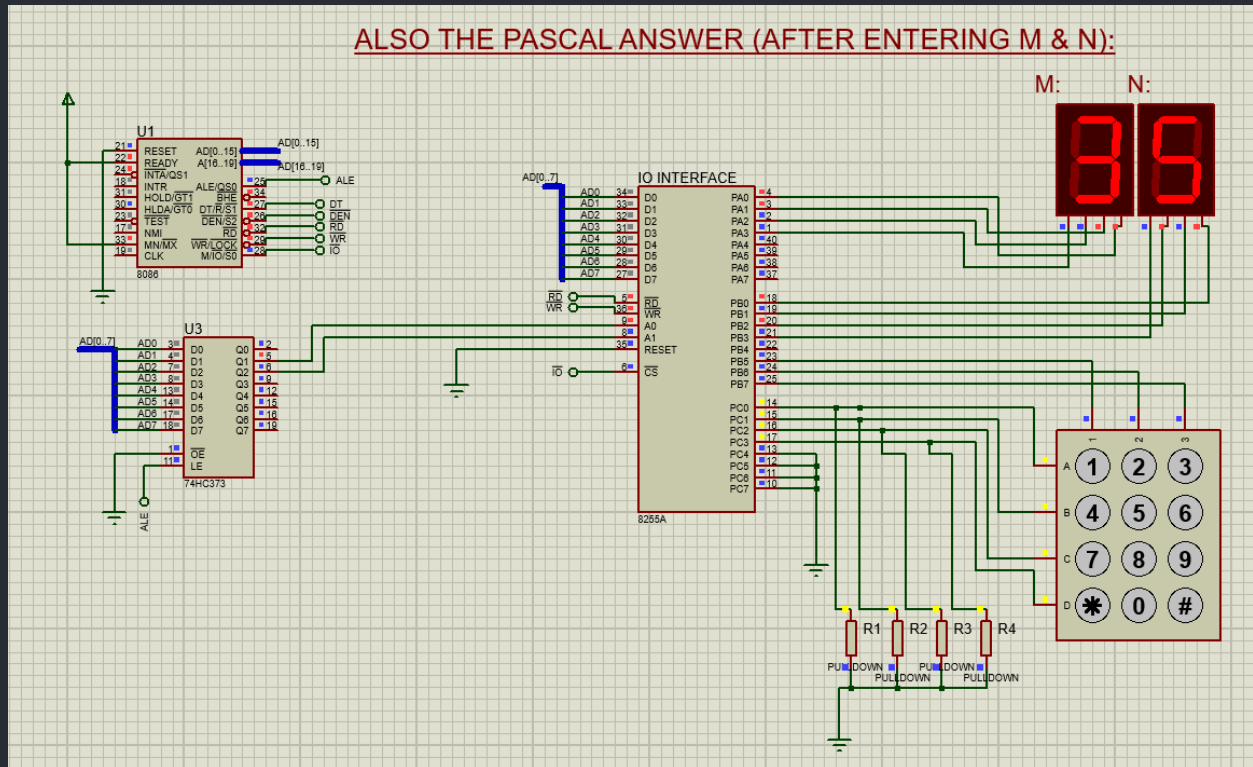


وارد کردن N:





محاسبه و خروجی (بعد از یک تاخیر که از قصد قرار داده شده):



## کد برنامه:

توضیحات به صورت کامنت و یا به صورت label و یا با اسم روتین ها مشخص شده است.

```
main.asm
1  .MODEL SMALL
2  .STACK 64
3  .DATA
4      ; --- IO 8285 PORTS AND CR ---
5      PORTA EQU 00H ; OUT - 1ST DIGIT
6      PORTB EQU 02H ; OUT - 2ND DIGIT
7      PORTC EQU 04H ; IN - KEYPAD
8      CNTRLREG EQU 06H
9      CNTRLBYTE EQU 10001001B
10     ; --- KEYPAD COLUMN ACTIVATIONS ON PORT B ---
11     FIRST_COL EQU 00100000B
12     SECOND_COL EQU 01000000B
13     THIRD_COL EQU 10000000B
14     ; -----
15     M DB ?
16     N DB ?
17     KPMN DB ?
18     ; -----
19     r DB 01H
20     x_fact DB ? ; IS USED TO CALCULATE factorial of x
21     n_fact DW ?
22     r_fact DW ?
23     n_r_fact DW ?
24     comb DB ? ; ANSWER OF Combination
25     OUTER_LOOP_COUNT_TEMP DW ?
26     ; -----
27  .CODE
28
29  MAIN    PROC FAR
30
31      MOV AX, @DATA
32      MOV DS, AX
33
34      CALL SET_CNTRLREG
35      CALL CLEAR_DIGITS
36
37      CALL TRACK_KEYPAD
38
39      CALL DELAY
40      CALL DELAY
```

```

42      ; CALL CLEAR_DIGITS
43      CALL CAL_KHPA
44      CALL BCD7SEG
45      CALL SHOW_NUMBER
46
47  ENDLESS:
48      JMP ENDLESS
49
50      ; MOV AH, 4CH
51      ; INT 21H
52  MAIN      ENDP
53
54
55  TRACK_KEYPAD PROC
56      MOV SI, 00H ; NUMBER OF NUMBERS TAKEN FROM KEYPAD
57
58  TRACKING_LOOP:
59      CMP SI, 02H
60      JE TRACKING_FINISH
61
62      CHECK_FIRST_COL:
63          MOV AL, FIRST_COL
64          OUT PORTB, AL
65          IN AL, PORTC
66          AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
67          JNZ HANDLE_FIRST_COL
68
69      CHECK_SECOND_COL:
70          MOV AL, SECOND_COL
71          OUT PORTB, AL
72          IN AL, PORTC
73          AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
74          JNZ HANDLE_SECOND_COL
75
76      CHECK_THIRD_COL:
77          MOV AL, THIRD_COL
78          OUT PORTB, AL
79          IN AL, PORTC

```

```

80      AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
81      JNZ HANDLE_THIRD_COL
82
83      JMP TRACKING_LOOP
84
85      HANDLE_FIRST_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN
86      CASE_ONE:
87          CMP AL, 01H
88          JNE CASE_FOUR
89          MOV BL, 01H
90          CALL SAVE_MN_TO_BX
91          JMP TRACKING_LOOP
92      CASE_FOUR:
93          CMP AL, 02H
94          JNE CASE_SEVEN
95          MOV BL, 04H
96          CALL SAVE_MN_TO_BX
97          JMP TRACKING_LOOP
98      CASE_SEVEN:
99          CMP AL, 04H
100         JNE CASE_STAR
101         MOV BL, 07H
102         CALL SAVE_MN_TO_BX
103         JMP TRACKING_LOOP
104      CASE_STAR:
105         JMP TRACKING_LOOP
106
107      HANDLE_SECOND_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN
108      CASE_TWO:
109          CMP AL, 01H
110          JNE CASE_FIVE
111          MOV BL, 02H
112          CALL SAVE_MN_TO_BX
113          JMP TRACKING_LOOP
114      CASE_FIVE:
115          CMP AL, 02H
116          JNE CASE_EIGHT
117          MOV BL, 05H
118          CALL SAVE_MN_TO_BX
119          JMP TRACKING_LOOP

```

```

120     CASE_EIGHT:
121         CMP AL, 04H
122         JNE CASE_ZERO
123         MOV BL, 08H
124         CALL SAVE_MN_TO_BX
125         JMP TRACKING_LOOP
126     CASE_ZERO:
127         CMP AL, 08h
128         JNE TRACKING_LOOP
129         MOV BL, 00H
130         CALL SAVE_MN_TO_BX
131         JMP TRACKING_LOOP
132
133     HANDLE_THIRD_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN
134     CASE_THREE:
135         CMP AL, 01H
136         JNE CASE_SIX
137         MOV BL, 03H
138         CALL SAVE_MN_TO_BX
139         JMP TRACKING_LOOP
140     CASE_SIX:
141         CMP AL, 02H
142         JNE CASE_NINE
143         MOV BL, 06H
144         CALL SAVE_MN_TO_BX
145         JMP TRACKING_LOOP
146     CASE_NINE:
147         CMP AL, 04H
148         JNE CASE_HASH
149         MOV BL, 09H
150         CALL SAVE_MN_TO_BX
151         JMP TRACKING_LOOP
152     CASE_HASH:
153         JMP TRACKING_LOOP
154     JMP TRACKING_LOOP
155 TRACKING_FINISH:
156     RET
157 TRACK_KEYPAD ENDP

```

```

160 SAVE_MN_TO_BX PROC ; GETS BL AS PARAMTER
161     CMP SI, 01H
162     JE SAVE_N ; FINISH IF THE PRESSED NUMBER IS N (LAST/2ND NUMBER)
163 SAVE_M: ; FIRST NUMBER
164     MOV DH, BL
165     JMP SAVE_FINISH
166 SAVE_N: ; SECOND NUMBER
167     MOV DL, BL
168     JMP SAVE_FINISH
169 SAVE_FINISH:
170     CALL SHOW_NUMBER
171     CALL DELAY
172     INC SI
173     RET
174 SAVE_MN_TO_BX ENDP
175
176
177 SHOW_NUMBER PROC ; GETS DX AS PARAMETER
178     PUSH AX
179     MOV AL, DH
180     OUT PORTA, AL
181     MOV AL, DL
182     OUT PORTB, AL
183     POP AX
184     RET
185 SHOW_NUMBER ENDP
186
187
188 CLEAR_DIGITS PROC ; MAKES ALL DIGITS ZERO
189     MOV DX, 0000H
190     CALL SHOW_NUMBER
191     RET
192 CLEAR_DIGITS ENDP
193
194
195 SET_CNTRLREG PROC
196     MOV AL, CNTRLBYTE
197     MOV DX, CNTRLREG
198     OUT DX, AL
199     RET

```

```

200 SET_CNTRLREG ENDP
201
202
203 DELAY PROC
204     PUSH CX
205     MOV CX, 0F000H
206     LOOP1:
207         NOP
208         LOOP LOOP1
209     POP CX
210     RET
211 DELAY ENDP
212
213
214 CAL_KHPA PROC
215     DEC DH
216     MOV M, DH
217     DEC DL
218     MOV N, DL
219
220     ; --- M! ---
221     MOV AH, M
222     MOV x_fact, AH ; x = M
223     MOV BL, AH
224     CALL FACT
225     MOV n_fact, AX
226     ; -----
227
228     ; --- N! ---
229     MOV DL, N
230     MOV x_fact, DL ; x = N
231     CALL FACT
232     MOV r_fact, AX
233     ; -----
234
235     ; --- (M-N)! ---
236     MOV DL, M ; DL = M
237     SUB DL, N ; DL = M - N
238     MOV x_fact, DL ; x = M - N
239     CALL FACT

```

```

240     MOV n_r_fact, AX
241     ; -----
242
243     ; ---  $M!/(N! (M-N)!)$  ---
244     MOV AX, n_fact
245     SUB DX, DX
246     DIV r_fact
247     SUB DX, DX
248     DIV n_r_fact
249     MOV comb, AL
250     ; -----
251
252     MOV DL, comb ; DL = comb (answer in binary)
253     RET
254 CAL_KHPA ENDP
255
256
257 FACT PROC ; Saves x! IN (DX, AX)
258
259     MOV OUTER_LOOP_COUNT_TEMP, CX
260
261     MOV CX, 01H ; COUNTER = 1
262     MOV AX, 01H ; AX = 1
263     MOV DX, 0000H ; CLEAR DX
264
265     BACK:
266     MOV BH, 00H
267     MOV BL, x_fact
268     CMP CX, BX
269     JA FINISH ; FINISH WHEN COUNTER > x
270     MUL CX ; (DX, AX) = CX * AX
271     INC CX ; COUNTER++
272     INC CX ; COUNTER++
273     LOOP BACK ; COUNTER--
274
275
276     FINISH:
277     MOV CX, OUTER_LOOP_COUNT_TEMP
278     RET
279

```



```

264
265 BACK:
266     MOV BH, 00H
267     MOV BL, x_fact
268     CMP CX, BX
269     JA FINISH ; FINISH WHEN COUNTER > x
270     MUL CX ; (DX, AX) = CX * AX
271     INC CX ; COUNTER++
272     INC CX ; COUNTER++
273     LOOP BACK ; COUNTER--
274
275
276 FINISH:
277     MOV CX, OUTER_LOOP_COUNT_TEMP
278     RET
279
280 FACT ENDP ; THE ANSWER IS STORED IN (DX, AX)
281
282
283 BIN2BCD PROC ; TAKES DL AS PARAMETER
284     XOR AH, AH
285     MOV AL, DL
286     MOV BL, 0AH ; AL / 10
287     DIV BL
288     MOV BH, AL ; AL = Quotient (Left Digit) | AL / 10
289     MOV BL, AH ; AH = Remainder (Right Digit) | AL % 10
290     RET
291 BIN2BCD ENDP ; RETURNS A 8-bit BCD IN BX
292
293
294 BCD7SEG PROC ; TAKES DL AS PARAMETER
295     CALL BIN2BCD
296     MOV DX, BX
297     RET
298 BCD7SEG ENDP ; RETURNS DX AS A BCD NO. FOR 7SEG
299
300
301 ; DO NOT REMOVE THIS LINE:
302     END MAIN

```

## شمای کلی مدار:



کد برنامه (توضیحات به صورت کامنت ذکر شده اند):

```

1  .MODEL SMALL
2  .STACK 64
3  .DATA
4      ; --- IO 8285 PORTS AND CR ---
5      PORTA EQU 00H ; OUT - 1ST DIGIT
6      PORTB EQU 02H ; OUT - 2ND DIGIT
7      PORTC EQU 04H ; IN - KEYPAD
8      CNTRLREG EQU 06H
9      CNTRLBYTE EQU 10001001B
10     ; --- IO 8253 PORTS AND CR ---
11     CLK0 EQU 08H ; OUT - 1ST DIGIT
12     CLK1 EQU 0AH ; OUT - 2ND DIGIT
13     CLK2 EQU 0CH ; IN - KEYPAD
14     CTRLREG EQU 0EH
15     CTRLBYTE EQU 00010110B
16     ; --- IO 8253 PORTS AND CR ---
17     ICW1 EQU 10H
18     ICW24 EQU 12H
19     ; --- KEYPAD COLUMN ACTIVATIONS ON PORT B ---
20     FIRST_COL EQU 00100000B
21     SECOND_COL EQU 01000000B
22     THIRD_COL EQU 10000000B
23     ; -----
24     DELAY_N DB 00FFH
25     ; -----
26     M DB ?
27     N DB ?
28     KPMN DB ?
29     ; -----
30     r DB 01H
31     x_fact DB ? ; IS USED TO CALCULATE factorial of x
32     n_fact DW ?
33     r_fact DW ?
34     n_r_fact DW ?
35     comb DB ? ; ANSWER OF Combination
36     OUTER_LOOP_COUNT_TEMP DW ?
37     SHOW_NUMBER_PAR DW ?
38     ; -----
39  .CODE
40

```

```
41 MAIN    PROC FAR
42
43     MOV AX, @DATA
44     MOV DS, AX
45
46     CALL SETUP_8255
47     CALL SETUP_8253
48
49     CALL INSERT_ISR_INTO_VECTOR_TABLE
50     CALL SETUP_8259
51
52
53     CALL CLEAR_DIGITS
54
55     CALL TRACK_KEYPAD
56
57     CALL CAL_KHPA
58     CALL BCD7SEG
59
60 LOOP_WINK:
61     MOV DH, 00H
62     MOV DL, comb
63     CALL BCD7SEG
64     CALL SHOW_NUMBER
65
66 ; --- DELAY SECTION ---
67     MOV CX, 0005H
68 LOOP_DELAY:
69     PUSH CX
70     CALL DELAYL
71     POP CX
72     LOOP LOOP_DELAY
73 ; -----
74     CALL CLEAR_DIGITS
75 ; --- DELAY SECTION ---
76     MOV CX, 0005H
77 LOOP_DELAY2:
78     PUSH CX
79     CALL DELAYL
```

```

79     CALL DELAYL
80     POP CX
81     LOOP LOOP_DELAY2
82     ; -----
83     JMP LOOP_WINK
84     ; CALL TRACK_KEYPAD_WINKING
85
86 ENDLESS:
87     JMP ENDLESS
88
89     ; MOV AH, 4CH
90     ; INT 21H
91 MAIN     ENDP
92
93 TRACK_KEYPAD_WINKING PROC
94     CALL CAL_KHPA
95     CALL BCD7SEG
96 WINK:
97     CALL SHOW_NUMBER
98     XOR CX, CX
99     MOV CL, DELAY_N
100 WINK_TRACKING_LOOP:
101     DEC CX
102     WINK_CHECK_FIRST_COL:
103         MOV AL, FIRST_COL
104         OUT PORTB, AL
105         IN AL, PORTC
106         AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
107         JNZ WINK_HANDLE_FIRST_COL
108
109     WINK_CHECK_SECOND_COL:
110         MOV AL, SECOND_COL
111         OUT PORTB, AL
112         IN AL, PORTC
113         AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
114         JNZ WINK_HANDLE_SECOND_COL
115
116     WINK_CHECK_THIRD_COL:
117         MOV AL, THIRD_COL
118         OUT PORTB, AL

```

```

109 WINK_CHECK_SECOND_COL:
110     MOV AL, SECOND_COL
111     OUT PORTB, AL
112     IN AL, PORTC
113     AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
114     JNZ WINK_HANDLE_SECOND_COL
115
116 WINK_CHECK_THIRD_COL:
117     MOV AL, THIRD_COL
118     OUT PORTB, AL
119     IN AL, PORTC
120     AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
121     JNZ WINK_HANDLE_THIRD_COL
122
123     CMP CX, 00H
124     JNE WINK_TRACKING_LOOP
125     JMP WINK_2
126
127 WINK_HANDLE_FIRST_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN
128     WINK_CASE_ONE:
129         CMP AL, 01H
130         JNE WINK_CASE_FOUR
131         MOV BL, 01H
132         CALL HANDLE_WINK
133         JNE WINK_TRACKING_LOOP
134     WINK_CASE_FOUR:
135         CMP AL, 02H
136         JNE WINK_CASE_SEVEN
137         MOV BL, 04H
138         CALL HANDLE_WINK
139         JNE WINK_TRACKING_LOOP
140     WINK_CASE_SEVEN:
141         CMP AL, 04H
142         JNE WINK_CASE_STAR
143         MOV BL, 07H
144         CALL HANDLE_WINK
145         JNE WINK_TRACKING_LOOP
146     WINK_CASE_STAR:
147         JNE WINK_TRACKING_LOOP
148

```

```

149 WINK_HANDLE_SECOND_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN
150     WINK_CASE_TWO:
151         CMP AL, 01H
152         JNE WINK_CASE_FIVE
153         MOV BL, 02H
154         CALL HANDLE_WINK
155         JNE WINK_TRACKING_LOOP
156     WINK_CASE_FIVE:
157         CMP AL, 02H
158         JNE WINK_CASE_EIGHT
159         MOV BL, 05H
160         CALL HANDLE_WINK
161         JNE WINK_TRACKING_LOOP
162     WINK_CASE_EIGHT:
163         CMP AL, 04H
164         JNE WINK_CASE_ZERO
165         MOV BL, 08H
166         CALL HANDLE_WINK
167         JNE WINK_TRACKING_LOOP
168     WINK_CASE_ZERO:
169         CMP AL, 08h
170         JNE WINK_TRACKING_LOOP
171         MOV BL, 00H
172         CALL HANDLE_WINK
173         JNE WINK_TRACKING_LOOP
174
175 WINK_HANDLE_THIRD_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN
176     WINK_CASE_THREE:
177         CMP AL, 01H
178         JNE WINK_CASE_SIX
179         MOV BL, 03H
180         CALL HANDLE_WINK
181         JNE WINK_TRACKING_LOOP
182     WINK_CASE_SIX:
183         CMP AL, 02H
184         JNE WINK_CASE_NINE
185         MOV BL, 06H
186         CALL HANDLE_WINK
187         JNE WINK_TRACKING_LOOP

```

```

190         JNE WINK_CASE_HASH
191         MOV BL, 09H
192         CALL HANDLE_WINK
193         JNE WINK_TRACKING_LOOP
194     WINK_CASE_HASH:
195         JNE WINK_TRACKING_LOOP
196     JNE WINK_TRACKING_LOOP
197
198 WINK_2:
199     CALL CLEAR_DIGITS
200     CALL DELAYL
201     CALL DELAYL
202     CALL DELAYL
203     CALL DELAYL
204
205     JMP WINK
206 WINK_TRACKING_FINISH:
207     RET
208 TRACK_KEYPAD_WINKING ENDP
209
210
211 HANDLE_WINK PROC
212     MOV DELAY_N, BL
213     RET
214 HANDLE_WINK ENDP
215
216 TRACK_KEYPAD PROC
217     MOV SI, 00H ; NUMBER OF NUMBERS TAKEN FROM KEYPAD
218
219 TRACKING_LOOP:
220     CMP SI, 02H
221     JE TRACKING_FINISH
222
223     CHECK_FIRST_COL:
224         MOV AL, FIRST_COL
225         OUT PORTB, AL
226         IN AL, PORTC
227         AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
228         JNZ HANDLE_FIRST_COL
229

```



```

230 CHECK_SECOND_COL:
231     MOV AL, SECOND_COL
232     OUT PORTB, AL
233     IN AL, PORTC
234     AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
235     JNZ HANDLE_SECOND_COL
236
237 CHECK_THIRD_COL:
238     MOV AL, THIRD_COL
239     OUT PORTB, AL
240     IN AL, PORTC
241     AND AL, 0FH ; CHECKS IF ANY BUTTON IN COL IS PRESSED
242     JNZ HANDLE_THIRD_COL
243
244     JMP TRACKING_LOOP
245
246 HANDLE_FIRST_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN
247     CASE_ONE:
248         CMP AL, 01H
249         JNE CASE_FOUR
250         MOV BL, 01H
251         CALL SAVE_MN_TO_BX
252         JMP TRACKING_LOOP
253     CASE_FOUR:
254         CMP AL, 02H
255         JNE CASE_SEVEN
256         MOV BL, 04H
257         CALL SAVE_MN_TO_BX
258         JMP TRACKING_LOOP
259     CASE_SEVEN:
260         CMP AL, 04H
261         JNE CASE_STAR
262         MOV BL, 07H
263         CALL SAVE_MN_TO_BX
264         JMP TRACKING_LOOP
265     CASE_STAR:
266         JMP TRACKING_LOOP
267
268 HANDLE_SECOND_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN

```

```

269     CASE_TWO:
270         CMP AL, 01H
271         JNE CASE_FIVE
272         MOV BL, 02H
273         CALL SAVE_MN_TO_BX
274         JMP TRACKING_LOOP
275     CASE_FIVE:
276         CMP AL, 02H
277         JNE CASE_EIGHT
278         MOV BL, 05H
279         CALL SAVE_MN_TO_BX
280         JMP TRACKING_LOOP
281     CASE_EIGHT:
282         CMP AL, 04H
283         JNE CASE_ZERO
284         MOV BL, 08H
285         CALL SAVE_MN_TO_BX
286         JMP TRACKING_LOOP
287     CASE_ZERO:
288         CMP AL, 08h
289         JNE TRACKING_LOOP
290         MOV BL, 00H
291         CALL SAVE_MN_TO_BX
292         JMP TRACKING_LOOP
293
294     HANDLE_THIRD_COL: ; SWITCH CASE FOR CORRESPONDING COLUMN
295     CASE_THREE:
296         CMP AL, 01H
297         JNE CASE_SIX
298         MOV BL, 03H
299         CALL SAVE_MN_TO_BX
300         JMP TRACKING_LOOP
301     CASE_SIX:
302         CMP AL, 02H
303         JNE CASE_NINE
304         MOV BL, 06H
305         CALL SAVE_MN_TO_BX
306         JMP TRACKING_LOOP
307     CASE_NINE:

```

```

310         MOV BL, 09H
311         CALL SAVE_MN_TO_BX
312         JMP TRACKING_LOOP
313     CASE_HASH:
314         JMP TRACKING_LOOP
315     JMP TRACKING_LOOP
316 TRACKING_FINISH:
317     RET
318 TRACK_KEYPAD ENDP
319
320
321 SAVE_MN_TO_BX PROC ; GETS BL AS PARAMTER
322     CMP SI, 01H
323     JE SAVE_N ; FINISH IF THE PRESSED NUMBER IS N (LAST/2ND NUMBER)
324     SAVE_M: ; FIRST NUMBER
325     MOV DH, BL
326     JMP SAVE_FINISH
327     SAVE_N: ; SECOND NUMBER
328     MOV DL, BL
329     JMP SAVE_FINISH
330     SAVE_FINISH:
331     MOV SHOW_NUMBER_PAR, DX
332     CALL SHOW_NUMBER
333     CALL DELAYL
334     INC SI
335     RET
336 SAVE_MN_TO_BX ENDP
337
338
339 SHOW_NUMBER PROC ; GETS DX AS PARAMETER
340     PUSH AX
341     MOV DX, SHOW_NUMBER_PAR
342     MOV AL, DH
343     OUT PORTA, AL
344     MOV AL, DL
345     OUT PORTB, AL
346     POP AX
347     RET
348 SHOW_NUMBER ENDP
349

```

```

351 CLEAR_DIGITS PROC ; MAKES ALL DIGITS ZERO
352     MOV SHOW_NUMBER_PAR, 0000H
353     CALL SHOW_NUMBER
354     RET
355 CLEAR_DIGITS ENDP
356
357
358 SETUP_8255 PROC
359     MOV AL, CNTRLBYTE
360     MOV DX, CNTRLREG
361     OUT DX, AL
362     RET
363 SETUP_8255 ENDP
364
365 SETUP_8253 PROC
366     MOV AL, CTRLBYTE
367     OUT CTRLREG, AL
368
369     MOV AL, 64H
370     OUT CLK0, AL
371     RET
372 SETUP_8253 ENDP
373
374 SETUP_8259 PROC
375
376     MOV AL, 00010011B           ; ICW1 : Edge Triggered - 1 Master - ICW4 Kullan
377     OUT ICW1, AL               ; SET InputControlWord1
378
379     MOV AL, 40H                ; ICW2 : 40h
380     OUT ICW24, AL              ; SET InputControlWord2
381
382     MOV AL, 03H                ; AEIO(Automatic Interrupt) = 1, PM(2 Acknowledge) = 1
383     OUT ICW24, AL              ; SET InputControlWord4
384
385     RET
386 SETUP_8259 ENDP
387 ;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
388

```

```

389 ;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
390 INSERT_ISR_INTO_VECTOR_TABLE PROC
391     CLI
392     XOR AX, AX
393     MOV ES, AX
394     MOV AL, 40H
395     MOV AH, 4
396     MUL AH
397     MOV BX, AX
398     LEA AX, INT_100_MSEC
399     MOV WORD PTR ES:[BX], AX
400     MOV AX, CS
401     MOV WORD PTR ES:[BX+2], AX
402     STI
403     RET
404 INSERT_ISR_INTO_VECTOR_TABLE ENDP
405
406
407 INT_100_MSEC PROC
408     INC DL
409     IRET
410 INT_100_MSEC ENDP
411
412
413 DELAYL PROC
414     PUSH CX
415     MOV CX, 0FFF0H
416     LOOP1:
417         NOP
418         LOOP LOOP1
419     POP CX
420     RET
421 DELAYL ENDP
422
423 DELAY PROC
424     PUSH DX
425     XOR DL, DL
426
427     STI

```

```

428  DELAY_ENDLESS:
429      CMP DL, DELAY_N
430      JNZ DELAY_ENDLESS
431
432      CLI
433      POP DX
434      RET
435  DELAY ENDP
436
437
438  CAL_KHPA    PROC
439      DEC DH
440      MOV M, DH
441      DEC DL
442      MOV N, DL
443
444      ; --- M! ---
445      MOV AH, M
446      MOV x_fact, AH ;  $x = M$ 
447      MOV BL, AH
448      CALL FACT
449      MOV n_fact, AX
450      ; -----
451
452      ; --- N! ---
453      MOV DL, N
454      MOV x_fact, DL ;  $x = N$ 
455      CALL FACT
456      MOV r_fact, AX
457      ; -----
458
459      ; --- (M-N)! ---
460      MOV DL, M ;  $DL = M$ 
461      SUB DL, N ;  $DL = M - N$ 
462      MOV x_fact, DL ;  $x = M - N$ 
463      CALL FACT
464      MOV n_r_fact, AX
465      ; -----
466

```

```

469     SUB DX, DX
470     DIV r_fact
471     SUB DX, DX
472     DIV n_r_fact
473     MOV comb, AL
474     ; -----
475
476     MOV DL, comb ; DL = comb (answer in binary)
477     RET
478 CAL_KHPA ENDP
479
480
481 FACT PROC ; Saves x! IN (DX, AX)
482
483     MOV OUTER_LOOP_COUNT_TEMP, CX
484
485     MOV CX, 01H ; COUNTER = 1
486     MOV AX, 01H ; AX = 1
487     MOV DX, 0000H ; CLEAR DX
488
489     BACK:
490         MOV BH, 00H
491         MOV BL, x_fact
492         CMP CX, BX
493         JA FINISH ; FINISH WHEN COUNTER > x
494         MUL CX ; (DX, AX) = CX * AX
495         INC CX ; COUNTER++
496         INC CX ; COUNTER++
497         LOOP BACK ; COUNTER--
498
499
500     FINISH:
501         MOV CX, OUTER_LOOP_COUNT_TEMP
502         RET
503
504     FACT ENDP ; THE ANSWER IS STORED IN (DX, AX)
505
506
507 BIN2BCD PROC ; TAKES DL AS PARAMETER
508     XOR AH, AH

```

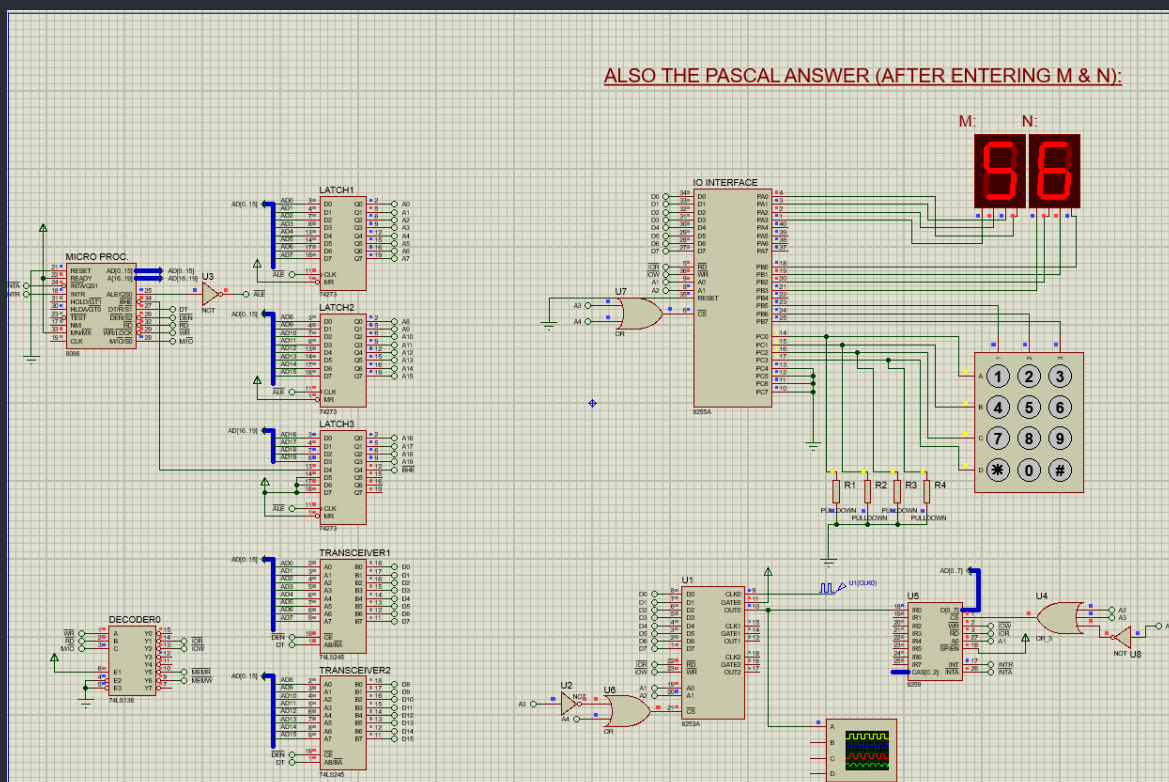
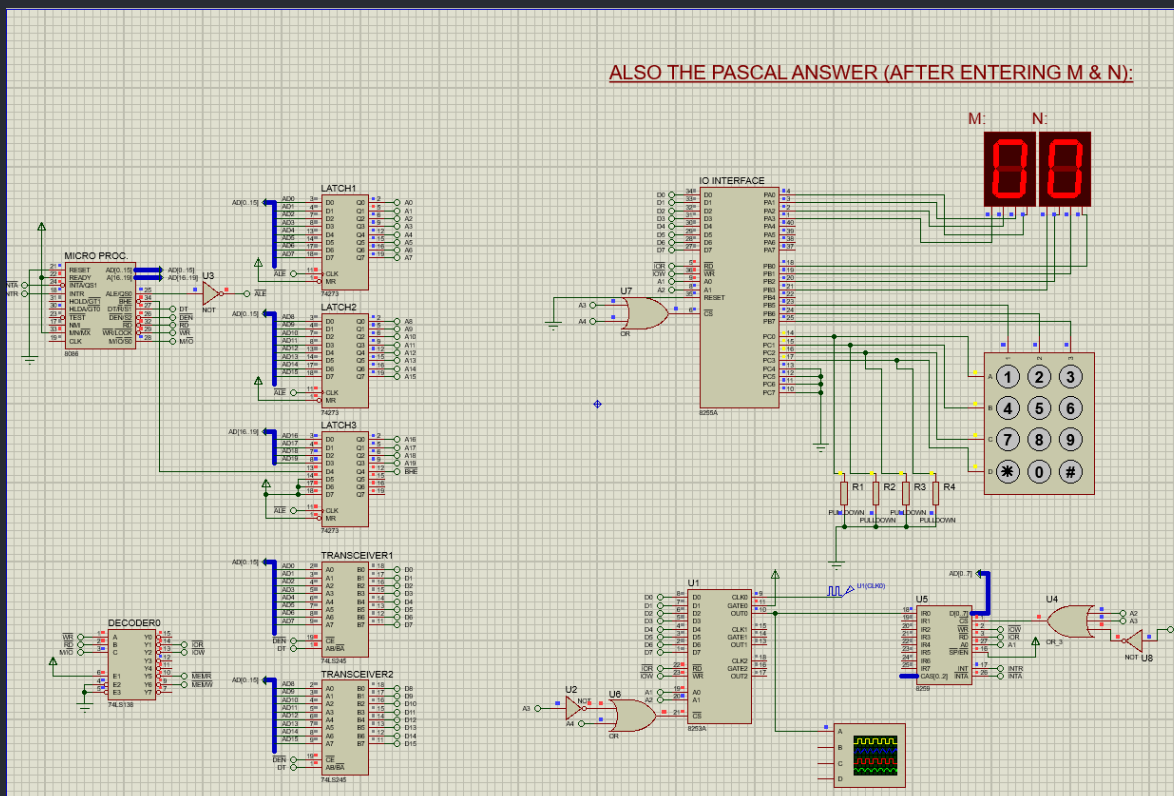
```

500 FINISH:
501     MOV CX, OUTER_LOOP_COUNT_TEMP
502     RET
503
504 FACT ENDP ; THE ANSWER IS STORED IN (DX, AX)
505
506
507 BIN2BCD PROC ; TAKES DL AS PARAMETER
508     XOR AH, AH
509     MOV AL, DL
510     MOV BL, 0AH ; AL / 10
511     DIV BL
512     MOV BH, AL ; AL = Quotient (Left Digit) | AL / 10
513     MOV BL, AH ; AH = Remainder (Right Digit) | AL % 10
514     RET
515 BIN2BCD ENDP ; RETURNS A 8-bit BCD IN BX
516
517
518 BCD7SEG PROC ; TAKES DL AS PARAMETER
519     CALL BIN2BCD
520     MOV SHOW_NUMBER_PAR, BX
521     RET
522 BCD7SEG ENDP ; RETURNS SHOW_NUMBER_PAR AS A BCD NO. FOR 7SEG
523
524
525 ; DO NOT REMOVE THIS LINE:
526     END MAIN

```



خروجی نمونه:



## پالس مربعی با پریود 100 ثانیه:

